

Yue CAO

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Paris, France



EXPERIENCE

Berger-Levrault

Feb. 2026 – Aug. 2026

NLP Research Intern

Boulogne, France

- **Project** : Intelligent Assistance Pack Generator for Offline Industrial Maintenance
- Prototyping a system that automatically generates compact assistance packs for offline tiny LLMs deployed on mobile devices, aimed at supporting maintenance technicians in disconnected environments under privacy, latency, and resource constraints
- Analyzing work orders to identify documentary needs and selecting the most relevant content from heterogeneous sources, including technical PDFs, documents, images, and intervention history
- Building, maintaining, and updating offline packs by preparing the chunking and generating the embeddings, indexes, and metadata required by the existing offline engine in its standard format, while taking into account tiny-LLM context constraints
- Optimizing pack quality through compression, deduplication, prioritization, and simple quality indicators to retain only the most useful content within strict size constraints
- Designing and testing an experimental RAG pipeline applied to the generated assistance packs, in order to evaluate their relevance, coherence and usability, and to guide the optimization of the produced contents, chunking, metadata and indexes

EDUCATION

INALCO – PluriTAL

Sept. 2025 – Present

Master's (M2) in Natural Language Processing

Paris, France

- **Coursework** : Machine Learning, Neural Networks, Language Modeling, Algorithmic Programming, Object-Oriented Programming, Web Technologies, Linguistics for Multilingual NLP

Université Paris Nanterre – PluriTAL

Sept. 2024 – June 2025

Master's (M1) in Natural Language Processing

Nanterre, France

- **Coursework** : Corpus Processing, Textual Statistics, Mathematics, Parallel Corpora, Treebanks and Parsing, Text Mining, Structured Documents, SQL

Heilongjiang International University

Sept. 2020 – June 2024

Bachelor of Arts in French Language and Literature

Heilongjiang, China

- **Dissertation** : Analyse des méthodes de traduction de métaphore dans les langues familières à l'exemple du feuilleton français *Dix pour Cent*

PROJECTS

RAG Pipeline for Question Answering

2026

- Built a complete RAG pipeline using ChromaDB and Gemini for question answering over the novel *My Brilliant Friend*
- Chunking, embedding, vector indexing, and prompt orchestration for grounded answers

Transformer Models for Classical Chinese

2026

- Trained BERT- and GPT-style Transformers from scratch for genre classification and classical Chinese text generation; experimented on historical prose and Tang poetry

Image Classification on CIFAR-10 with CNN

2026

- Trained and evaluated a CNN on CIFAR-10 using PyTorch, and improved performance through architecture tuning and training refinements

- Applied data augmentation, shuffled mini-batches, and ReLU-based optimization strategies to improve classification results

Urban Traffic Forecasting in Paris Using Traffic and Weather Time Series

2026

- Built a short-term urban traffic forecasting pipeline using Paris Open Data traffic counts and Open-Meteo weather variables
- Preprocessed raw JSON traffic data, aligned traffic and weather time series, and engineered temporal and seasonal features with cyclic encoding
- Trained and compared a historical baseline, HistGradientBoosting, LSTM, and GRU models for multivariate time-series forecasting
- Developed a Streamlit application to load trained models and generate predictions without retraining

SKILLS

Programming : Python, SQL, Bash/Shell, HTML, CSS, XML

Libraries and frameworks : pandas, NumPy, scikit-learn, Keras, TensorFlow, NLTK, spaCy, Stanza, Hugging Face Transformers

Machine Learning and Deep Learning : Word2Vec, CNN, RNN, LSTM/GRU, Transformer architectures (BERT, GPT)

Data and document processing : corpus processing, structured documents, information extraction, OCR, metadata, indexing

NLP and machine learning tasks : RAG, sentiment analysis, text classification, image classification, topic modeling, text generation, information extraction, OCR, ASR, TTS, speech synthesis

Tools : VS Code, Jupyter, Overleaf, Oxygen XML, MySQL, Praat, Tesseract, Unitex, marimo, Neo4j, NooJ

Languages : Chinese (native), French (fluent / professional), English (fluent / professional)